



International School

Capstone Project 2

CMU-SE 451

C2SE.03

PROJECT PROPOSAL

Version 1.0

Date: August 20, 2026

EduShare Vietnam - Management system for receiving, coordinating and distributing educational resources for students in difficult circumstances.

Submitted by

Phuoc, Nguyen Thanh

Chi, Do Tran Uyen

Bao, Nguyen Cong

China, Ngo Quoc

Tri, Vo Van Duc

Approved by MSc. Huy, Truong Dinh

Proposal Review Panel Representative:

Name Signature Date

Capstone Project 2- Mentor:

Name

Signature

Date

Da Nang, 08/2026

PROJECT INFORMATION

Project acronym	EduShare Vietnam		
Project Title	Management system for receiving, coordinating and distributing educational resources for students in difficult circumstances.		
Start Date	20 Aug, 2026	End Date	11 May, 2026
Lead Institution	International School, Duy Tan University		
Project Mentor	MSc Huy, Truong Dinh Email: huy.truongdinh@gmail.com/ huytd@duytan.edu.vn Tel: 0982132352		
Scrum master / Project Leader & contact details	Phuoc, Nguyen Thanh Email: nguyenthanhphuoc24082000@gmail.com Tel: 0838510345		
Partner Organization			
Project Web URL			
Team members	Name	Email	Tel
28211100636	Phuoc, Nguyen Thanh	nguyenthanhphuoc24082000@gmail.com	0838510345
28200204644	Chi, Do Tran Uyen	uyenchido1231@gmail.com	0328321205
28214900367	Bao, Nguyen Cong	congbaonguyen45@gmail.com	0914962779
25211415118	China, Ngo Quoc	quoctrung4real@gmail.com	0899391923
28211101270	Tri, Vo Van Duc	ninjaschool37@gmail.com	0706007338

DOCUMENT APPROVALS

The following signatures are required for approval of this document.

MSc Huy, Truong Dinh <i>Mentor</i>	Signature	Date
Phuoc, Nguyen Thanh Student ID: 28211100636 <i>Scrum Master</i>	Signature	Date
Chi, Do Tran Uyen Student ID: 28200204644 <i>Team Member</i>	Signature	Date
Bao, Nguyen Cong Student ID: 28214900367 <i>Team Member</i>	Signature	Date
China, Ngo Quoc Student ID: 25211415118 <i>Team Member</i>	Signature	Date
Tri, Vo Van Duc Student ID: 28211101270 <i>Team Member</i>	Signature	Date

DOCUMENT NAME

Document Title	Project Proposal Document		
Author(s)	Nguyen Thanh Phuoc		
Role	Product Owner, Team Member, Scrum Master		
Date	August 20, 2026	File name:	C2SE.03_ProjectProposal_ver 1.0.docx

REVISION HISTORY

Version	Date	Comments	Author
1.0	25 Aug, 2026	Initial Release	All Member

Table of Contents

1. Project Title.....	1
2. Team members.....	1
3. Supervisor	2
4. Problem Statement	2
5. Existing Solutions	3
6. Objectives and Scope.....	4
6.1. Objectives.....	4
6.2. Scope.....	4
7. Key Features	5
7.1. Key Features.....	5
7.2. Quality Attribute	6
8. Constraints and Assumptions.....	7
8.1. Constraints.....	7
8.2. Assumptions.....	8
9. Target Users	8
10. Technology Stack	9
11. Methodology & Development Plan	9
11.1. Methodology	9
11.2. Development Plan	10
12. System Architecture Overview	11
13. Potential Risks and Mitigation Strategies	12
14. Expected Outcomes / Deliverables	13
15. References.....	15
16. Attached “DESCRIPTION OF PRODUCT REQUIREMENTS”	16
16.1. Brief description of product ideas (less than 7 statements)	16
16.2. Requirements.....	16

1. Project Title

EduShare Vietnam - Management system for receiving, coordinating and distributing educational resources for students in difficult circumstances.

2. Team members**Table 1.** *Team members*

TEAM MEMBER	
Phuoc, Nguyen Thanh	Email: Student ID: 28211100636 Role: Description:
Chi, Do Tran Uyen	Email: Student ID: Role: Description:
Bao, Nguyen Cong	Email: Student ID: Role: Description:
Trung, Ngo Quoc	Email: Student ID: Role: Description:
Tri, Vo Van Duc	Email: Student ID: Role: Description:

3. Supervisor

Table 2. *Supervisor*

SUPERVISOR	
Huy, Truong Dinh	Email: huy.truongdinh@gmail.com/ huytd@duytan.edu.vn Degree: MSc - Master of Science Tel: 0982132352 Faculty: International School - CMU

4. Problem Statement

Every year in Vietnam, a large amount of educational resources — textbooks, uniforms, backpacks, personal computers, laptops, tablets, bicycles, desks, chairs, and other learning tools — are still in good use but no longer used by the original owners, especially after each school year or when students change grades.

Meanwhile, many students in remote and disadvantaged areas still lack basic learning materials. Many children do not have enough textbooks or uniforms at the beginning of the school year, many schools do not have enough computers to teach Information Technology, and many students cannot participate in online learning simply because they do not have a computer or tablet at home.

Currently, the majority of donation activities to connect the above two situations still take place manually and on a small scale, through Excel spreadsheets, Zalo groups, Facebook posts or paperwork shared informally between individuals, former students, teachers and volunteer groups. There is no digital platform that systematically manages the flow of resources from those who no longer need them, to the students and schools who need them most.

From a donor perspective, individuals, parents, alumni, and businesses who want to donate unwanted educational resources often do not have a centralized, trusted channel to register donations, track how resources are used, and see the impact they make. This reduces the incentive to make long-term, continuous contributions.

From the perspective of receiving schools and students, schools, teachers and volunteer organizations have difficulty communicating their actual needs - what kind of resources are needed, how many, for how many students - to potential donors. Requests posted on social networks quickly get lost in the news flow, and the resources received often do not match the actual needs.

From a coordination perspective, there is no system in place to track which resources have been donated, where they are being stored, what their status is, and to which school or student they have been allocated. As a result, the distribution of resources is often uneven: one place has an excess of donated books while another has a shortage, goods sit undistributed for long periods of time, and the entire process lacks transparency.

To solve the above problems, our team proposed to build EduShare Vietnam an educational resource management platform, not a simple charity website.

The system builds a connected ecosystem:

- Sponsors — individuals, parents, alumni, businesses and publishers
- Receiving unit — school, education department, youth union, student union
- Volunteers and coordinators — in charge of receiving, checking, sorting, storing and handing over resources
- Students and schools benefit

The platform allows donors to register and track donations online, supports coordinators in receiving, auditing, sorting and managing resources through a centralized warehouse system, allows schools and students to submit and prioritize support requests, and applies a structured data reconciliation mechanism to properly allocate resources between surplus and shortage.

The proposed system aims to reduce waste of usable educational resources, increase transparency and efficiency in coordinating donation activities, and contribute to creating more equitable learning opportunities for disadvantaged students through a centralized digital platform.

5. Existing Solutions

Table 3. *Survey Overview*

Criteria	Traditional donation	Social network donation group	Existing charity / donation platforms	EduShare Vietnam (System recommendation system)
Centralized donation registration	X	Optional	Optional	✓
Warehouse and inventory management resources	X	X	Optional	✓
Classify & evaluate item condition	X	X	X	✓
Track items using QR/Barcode	X	X	X	✓
Manage support requests (school/student)	X	X	Optional	✓

Criteria	Donate transmission system	Donation group contribute online social	Foundation from charity / donation existing contribution	EduShare Vietnam (Hệ recommendation)
Compare needs and allocate resources	X	X	X	✓
Manage seasonal/regional campaigns	Optional	X	Optional	✓
Assign volunteer tasks	Optional	X	X	✓
Track shipping and handover	X	X	X	✓
Statistics and social impact reporting	X	X	Optional	✓
Multi-role dashboard & reporting	X	X	X	✓

Our system overcomes the limitations of existing solutions through the cải tiến sau:

- Centralized donation registration and coordination:** Donors can register to donate online instead of relying on sporadic posts or personal messages. This Replace manual, fragmented management with a single platform that holds data Clear and traceable donations.
- Classify and trace resources by status:** Each resource is donated are inspected, classified and assigned an identification code so that condition can be tracked, storage location and distribution history from receipt to delivery to the recipient enjoy.
- Centralized warehouse management:** System for managing inventory levels, storage locations and status each item (pending, inspected, ready to deliver, delivered) on one or multiple warehouses, reducing the risk of being lost or left behind for a long time.
- Compare needs and allocate according to priority:** Schools and students can Submit a support request describing your specific needs. The system reconciles existing inventory with the pending requests and proposed allocation options based on urgency, location location and condition of goods, instead of relying on manual decisions.
- Seasonal and regional campaign management:** Platform to support organizing activities donate into structured campaigns (e.g. relay campaigns to

school), each campaign has a goal, time, list of items to donate and dedicated implementation progress — something social media teams cannot provide.

- **Volunteer coordination:** Volunteers can be assigned tasks specifically receiving, sorting, packaging or transporting resources, and donating their contributions can be recognized — something that is not supported in fundraising activities traditional contributions.
- **Transparency and social impact reporting:** Donors and partner organizations have can know how much was donated, where the goods went, and how many students, schools are supported, thereby building long-term trust and encouraging contributions frequent.

6. Objectives and Scope

6.1. Objectives General Objective:

General goal:

The overall goal of the project is to build an educational resource management platform (Education Resource Management Platform) connects sponsors, receiving units/schools, and volunteers volunteer/coordinator and beneficiary students.

The system aims to provide a centralized solution to coordinate donation activities Collect, classify, store and redistribute educational resources, reducing waste of remaining resources usability, improve the transparency and efficiency of the donation process, and promote promote more equitable learning opportunities for disadvantaged students through digital technology.

Specific goals (according to the SMART model):

- **Specific:** The project builds a centralized platform that allows sponsors to post Sign donations of educational resources, support coordinators in receiving, classifying and managing Manage resources in a digital repository system, allowing schools and students to submit requests request support, and assist the coordinator in allocating resources to appropriate requests.
- **Measurable:** The system delivers measurable results through allowing users to go through the entire process from donation to distribution, including donation registration, receiving and sorting items, inventory management, shipping and approve support requests, allocate resources, manage campaigns and volunteers, All are reflected through the reporting dashboard.
- **Achievable:** The project can be implemented by integrating Modern web application technology, backend services and relational database management systems.

The development team will build a complete prototype product including:

Modules for donors, coordinators/administrators, receiving units and volunteers during project implementation.

●**Relevant:** The project directly solves the existing problem of resources

Education is wasted while students' needs are not met or adopted

building a digital ecosystem for donation coordination activities. Contribution system part of promoting equity in access to education, reducing waste of resources and creating community-based positive social impact.

Time-bound: The project will be developed and completed within a period of time of the Capstone module, including the stages of requirements analysis, system design, and development Deploy, test, deploy and complete final documentation.

6.2. Scope

Includes:

Donor:

Sponsor support system with the following functions:

- Register an account and log in to the system.
- Manage personal or organizational profile information.
- Register donations, including resource type, amount, status and image proof.
- Choose the appropriate delivery address and time.
- Track the status of your registered donation.
- View donation history and total donation value.
- View the list of campaigns and choose to contribute to a specific campaign.
- See reports on where donated resources are delivered to.

Đơn vị tiếp nhận / Trường học / Người thụ hưởng:

The system supports schools, teachers and beneficiary organizations with the following functions:

- Register an account for the receiving unit.
- Submit a support request, specifying the resource type, quantity, and reason for support.
- Specify the urgency of the request.
- Track the status of submitted requests (awaiting approval, approved, allocated, discussed deliver).
- Get notified when requests are approved or resources are allocated.

- Confirm receipt of allocated resources.

Volunteer / Coordinator:

The system supports volunteers and coordinators with the following functions:

- Confirm the donation reception schedule with the sponsor.
- Record the quantity and actual status of received resources.
- Classify resources by category and assign traceable identifiers (QR/Barcode).
- Inventory management, including warehousing, warehousing and transferring.
- Review and prioritize support requests for review.
- Compare approved requests with existing inventory and propose allocation plans.
- Coordinate transportation and confirm handover to the receiving unit.
- Assigned and reported on volunteer tasks (receiving, packing, transport, ...).

Campaign Management:

The system supports creating and managing donation campaigns with the following functions:

- Create a campaign with a name, goal, time, and list of items to donate góp.
- Track quantities received and delivered against campaign goals.
- Generate campaign results reports.

Administrator:

The system supports administrators with the following functions:

- Manage user accounts and assign permissions by role.
- Verify donor, school and organization accounts.
- Monitor warehouse activities and campaigns across the system.
- Create dashboards and reports on donations, distributed resources, numbers students/schools supported and estimated value of resources.

Excludes:

The project does not include:

- Operate your own fleet of vehicles/transportation services (coordination system with volunteers employees or partner transport units instead of operating the fleet themselves).
- Detailed financial fundraising functions and online payment processing for the cash donations.

- Physical repair or refurbishment services for damaged electronic devices.
- AI-based automatic pricing or image classification of items (okay identified as a potential future expansion direction).
- Transparent ledger based on blockchain technology (defined as open expand potential in the future).
- Replace official state-run charity fund management systems.

7. Key Features

7.1. Key Features

Table 4. *Functional Requirements*

ID	Requirement Name	Description
F01	Manage accounts and roles	The system supports account creation, authentication, profile management, and role-based access permissions for sponsors, schools, employees, volunteers, and administrators.
F02	Manage educational resource donations	The system allows donors to register educational resources for donation, provide resource information, upload images, and track donation status.
F03	Check and classify resources	The system allows receiving staff to check donated resources, assess condition, classify resources, and update resource status.
F04	Warehouse and inventory management	The system supports inventory management, including warehousing, warehousing, resource location tracking, and resource status monitoring.
F05	Manage support requests	The system allows schools or organizations to submit requests for educational resources and track request status.
F06	Intelligent resource allocation	The system supports resource allocation based on available inventory, request priority, resource status, and other relevant factors.
F07	Track resources using QR codes	The system supports creating and scanning QR codes to track the movement of resources during the stages of reception, storage, and distribution and handover.
F08	Volunteer management and distribution	The system allows volunteers to receive assigned tasks, update transport progress and confirm resource deliveries.

ID	Requirement Name	Description
F08	Volunteer management and distribution	The system allows volunteers to receive assigned tasks, update transport progress and confirm resource deliveries.
F09	Manage notifications	The system provides notifications regarding donation status, support requests, allocation decisions, and distribution activities.
F10	Dashboard and social impact tracking	The system provides dashboards and reports to monitor resources, supported schools, beneficiaries, delivery activities and social impact.
F11	Campaign management	The system allows administrators or organizations to create and manage educational resource support campaigns.
F12	Administrative management	The system allows administrators to manage users, resources, requests, allocations, and translations, system operations and reporting.

As a Sponsor, users can:

- Register and log in to the system.
- Manage personal account information.
- Sign up for educational resources to donate.
- Provides detailed information about donated resources.
- Upload images of educational resources.
- Specifies the quantity and status of the donated resource.
- Track the status of donation requests.
- View donation history.
- Receive notifications about donation processing.
- See information about the impact of your contributions.

As a School or Educational Institution, users can:

- Register and manage organizational accounts.
- Send verification information when necessary.
- Manage organizational profile information.
- Submit a request for support with educational resources.
- Specifies the type and quantity of resources needed.
- Provides information about the number of students or beneficiaries.
- Indicates the urgency or priority of the support request.
- Tracks the status of submitted requests.

- Receive notifications about allocation decisions.
- Confirm receipt of distributed resources.
- View support request history.

With the role of Receptionist, users can:

- Log in with your employee account.
- View incoming donation requests.
- Verify donation resource information.
- Check the physical condition of educational resources.
- Classify resources by type and status.
- Approve or deny inappropriate resources.
- Update resource status after checking.
- Create or assign a QR code to an accepted resource.
- Move approved resources to the warehouse management process.

With the Warehouse Worker role, users can:

- Log in with the warehouse employee account.
- View educational resources currently in the repository.
- Record resource warehousing activities.
- Record resource release activities.
- Update resource location.
- Monitor inventory quantity.
- Monitor resource health and status.
- Identify resources that have been stored for a long time.
- Support inventory checking and reporting.

As a Volunteer, users can:

- Register and log in to the system.
- Manage personal profile information.
- View available or assigned volunteer tasks.
- Accept assigned transportation or distribution tasks.
- View resource pickup and delivery information.
- Update shipping progress.
- Confirm resource has been retrieved.
- Confirmation of successful delivery or handover.

- View volunteer activity history.

As a Resource Allocation System, the system provides:

- Connection between existing educational resources and support requests.
- Check inventory before allocating resources.
- Recommend resources based on requirements.
- Supports priority-based allocation.
- Consider the quantity and status of resources.
- Suggest allocation to admin or coordinator.
- Track allocation status.
- Supports minimizing unnecessary shortages or surpluses of resources.

With the Administrator role, users can:

- Log in with an administrator account.
- Manage user accounts and roles.
- Approve and manage school or organization accounts.
- Manage educational resource catalog.
- Monitor donation activities.
- Monitor resource testing activities.
- Manage warehouse information.
- Review support requests.
- Approve or manage resource allocation decisions.
- Manage educational support campaigns.
- Monitor volunteer activities and distribution.
- Manage system notifications.
- Monitor entire system activity.
- Create reports related to:
 - +Total number of resources donated.
 - +Current inventory.
 - +Allocated resources.
 - + Các trường học hoặc tổ chức được hỗ trợ.
 - + Số lượng người thụ hưởng.
 - +Donation activities.
 - +Resource allocation activities.

+ Tác động xã hội tổng thể.

7.2. Non – Functional Requirements

Table 5. *Non – Functional Requirements*

ID	Request	Mô tả
NFR01	Performance	The system needs to respond effectively under normal usage conditions.
NFR02	Usability	The system needs to provide an intuitive interface for all user roles.
NFR03	Availability	The system needs to be accessible via a web browser with stable operation.
NFR04	Reliability	Resource and allocation data must be handled correctly.
NFR05	Security	User information must be protected through authentication and authorization.
NFR06	Scalability	The system needs to support future expansion.
NFR07	Maintainability	The system needs to use a modular architecture.
NFR08	Integration capabilities	The system needs to support future integration with maps, notifications, and external services.

8. Constraints and Assumptions

8.1. Constraints

Table 6. *Constraints*

Constraints	Description
Time constraints	The project must be completed within the time frame Capstone course time.
Technical constraints	The team has limited resources for scale infrastructure big.
Data Dependency	Resource data and requirements depend on information provided by users.
Third-party dependencies	Some services may depend on QR,storage cloud or notification API.
Security constraints	Personal and organizational information must be protected.
Logistical constraints	The project does not operate a transportation network independence.
Budget constraints	The project has a limited budget for cloud and Paid APIs.

8.2. Assumptions

The project assumes that:

- User has an internet-connected device.
- Sponsor provides accurate information.
- Schools submit valid support requests.
- Agent verifies accurate resource status.
- Volunteers update accurate delivery status.
- The system has enough sample data for demonstration.

9. Target Users

Table 7. *Target Users*

Target Users	Description
Sponsor	- Create an account and log in to the system. - Manage personal account information. - Sign up for educational resources to donate. - Provide detailed information about donated resources. - Upload images of educational resources. - Specify the quantity and status of resources. - Track donation status. - View donation history. - Receive notifications about donation processing and fund status. - See information about the impact of your contributions.
School / Group educational institution	- Create and manage organizational accounts. - Manage organizational profile information. - Submit a request for support regarding educational resources. - Specify the type and quantity of resources needed. - Provide information about the number of students or beneficiaries. - Indicates the urgency or priority of the support request. - Track request status. - Get notified about allocation decisions. - Confirm receipt of distributed resources.

Target Users	Description
Staff continued receive	- Log in with your employee account. - View incoming donation registrations. - Verify donated resource information. - Check the physical condition of educational resources. - Classify resources by type and status. - Approve or reject inappropriate resources. - Update resource status after checking. - Create or assign QR codes to approved resources. - Transfer approved resources to the management process kho.
Warehouse employee	- Log in with your warehouse employee account. - View educational resources currently in stock. - Record warehousing activities. - Record warehouse delivery activities. - Update resource locations. - Monitor inventory quantity. - Monitor resource health and status. - Support inventory checking and reporting. - View resources stored for a long time.
Volunteer	- Create an account and log in to the system. - Manage personal profile information. - View existing or assigned volunteer tasks. - Accept assigned shipping or distribution tasks. - View resource pickup and delivery information. - Update shipping progress. - Confirm resource has been taken. - Confirm successful delivery or handover. - View volunteer activity history.

Target Users	Description
Administrator	- Manage user accounts and roles. - Manage information about sponsors, schools, employees and volunteers volunteer. - Verify school or organization accounts when necessary. - Manage educational resource catalog. - Supervise donation activities. - Supervise resource testing activities. - Manage warehouse information. - Review support requests. - Approve or manage resource allocation decisions. - Manage educational support campaigns. - Supervise volunteer activities and distribution. - Manage system notifications. - Monitor all system operations. - Create reports on donated resources, inventory, and spent resources distribution, schools supported, beneficiaries and impact xã hội.

10. Technology Stack

•Frontend:

- React
- Tailwind CSS

•Backend:

- NestJS
- JWT
- RBAC
- Prism

•Database (DBMS):

- PostgreSQL

•Storage:

- Cloudinary

•Notification:

- Email

•Deployment:

- Docker

•QR: ○ QR Code

•Dashboard

○Chart.js

11. Methodology & Development Plan

11.1. Methodology

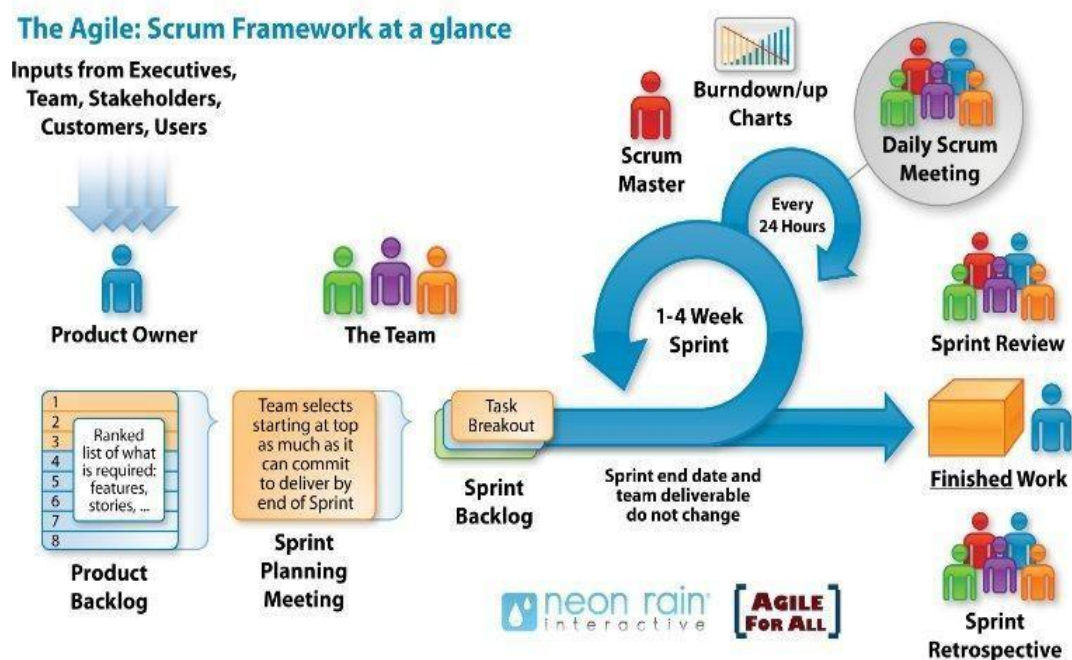


Figure 1. General SCRUM Process

- Scrum is an iterative and incremental agile software development framework for managing software projects and product or application development.
- Scrum focuses on project management institutions where it is difficult to plan ahead.
- Mechanisms of empirical process control, where feedback loops that constitute the core management technique are used as opposed to traditional command-and-control management.
- Its approach to planning and managing projects is by bringing decision-making authority to the level of operation properties and certainties.

11.2. Development Plan

Table 8. Project Timeline

STT	Job Name / Sprint Phase	Time quantity (Date)	Capture date first	End date end	Block quantity (Hours effort)
-----	-------------------------	----------------------	--------------------	--------------	-------------------------------

1.0	Inception Phase	9	20/08/2026	28/08/2026	18
1.1	Project kick-off meeting, unifying ideas and assigning roles	1	20/08/2026	20/08/2026	2
1.2	Survey practical problems, analyze educational resource management operations	2	21/08/2026	22/08/2026	4
1.3	Write, complete and protect the Proposal Document (Proposal Document)	6	23/08/2026	28/08/2026	12
2.0	3 Sprint Implementation & Development Phase (Development Phase)	60	29/08/2026	27/10/2026	1.432
2.1	Sprint 1: Architecture, RBAC Authentication, Donor Portal & QR Code Onboarding	21	29/08/2026	18/09/2026	490
2.1.1	Sprint 1 Planning & Product Backlog Decomposition	1	29/08/2026	29/08/2026	20
2.1.2	Configure Docker, design PostgreSQL and Redis database schema	3	30/08/2026	01/09/2026	60
2.1.3	Develop RBAC Authentication and Authorization module (Spring Security / JWT)	4	02/09/2026	05/09/2026	80
2.1.4	Designed Figma UI/UX Wireframe and built Web/Mobile Base Navigation framework	4	06/09/2026	09/09/2026	80
2.1.5	Build a Donor Portal: Register to donate resources & Download proof photos	3	10/09/2026	12/09/2026	70
2.1.6	Goods Receiving Module & Tool to generate QR/Barcode codes to identify items	4	13/09/2026	16/09/2026	90
2.1.7	Unit testing, bug fixing, Review and Retrospective Meeting Sprint 1	2	17/09/2026	18/09/2026	90
2.2	Sprint 2: Warehouse Management, Quality Inspection, Repairs & School Needs	20	19/09/2026	08/10/2026	480
2.2.1	Sprint 2 planning & Clarifying warehouse inspection business requirements	1	19/09/2026	19/09/2026	20
2.2.2	Multi-point Warehouse Management Subsystem: Shelf location, Import - Export - Warehouse transfer	3	20/09/2026	22/09/2026	75
2.2.3	Build Inspection Form Quality determination & classification (Books, Uniforms, Electronics)	3	23/09/2026	25/09/2026	75
2.2.4	Equipment Repair & Refurbishment Management Module (Replace components, install	3	26/09/2026	28/09/2026	70

	OS)				
2.2.5	Support Needs Registration Portal for Schools and Teachers	3	29/09/2026	01/10/2026	80
2.2.6	Priority Scoring and Evaluation Algorithm	3	02/10/2026	04/10/2026	75
2.2.7	Thematic Campaign Management Module ("Support to school", "Computers for children")	2	05/10/2026	06/10/2026	45
2.2.8	Sprint 2 Integration Testing, Source Code Inspection and Retrospective Meeting	2	07/10/2026	08/10/2026	40
2.3	Sprint 3: AI Coordination Algorithm, Volunteer Management & ESG Dashboard	19	09/10/2026	27/10/2026	462
2.3.1	Sprint 3 Planning & Multi-Warehouse Coordination Algorithm Specification	1	09/10/2026	09/10/2026	20
2.3.2	Building Python Microservices: Warehouse Control Algorithm supply and demand	4	10/10/2026	13/10/2026	100
2.3.3	Volunteer Management Subsystem: Shift assignment, Task assignment & QR attendance	3	14/10/2026	16/10/2026	75
2.3.4	Shipping & Handover Process: Electronic Signature, Photo Confirmation (PoD)	3	17/10/2026	19/10/2026	72
2.3.5	Lookup Portal Transparent item journey & Export Contribution Certificate	2	20/10/2026	21/10/2026	50
2.3.6	Building a Dashboard for Social Impact Measurement and Financial Reporting	3	22/10/2026	24/10/2026	80
2.3.7	Comprehensive System Testing (End-to-End), Load Testing and Retrospective Meeting	3	25/10/2026	27/10/2026	65
3.0	System Testing, Acceptance Error Fixing & UAT	10	28/10/2026	06/11/2026	30
4.0	Project Summary Meeting & Document Packaging	1	07/11/2026	07/11/2026	15
5.0	Release Final Version & Protect Project	1	08/11/2026	08/11/2026	30
Tổng	Toàn bộ Vòng đời Dự án	81	20/08/2026	08/11/2026	1.525

12. System Architecture Overview

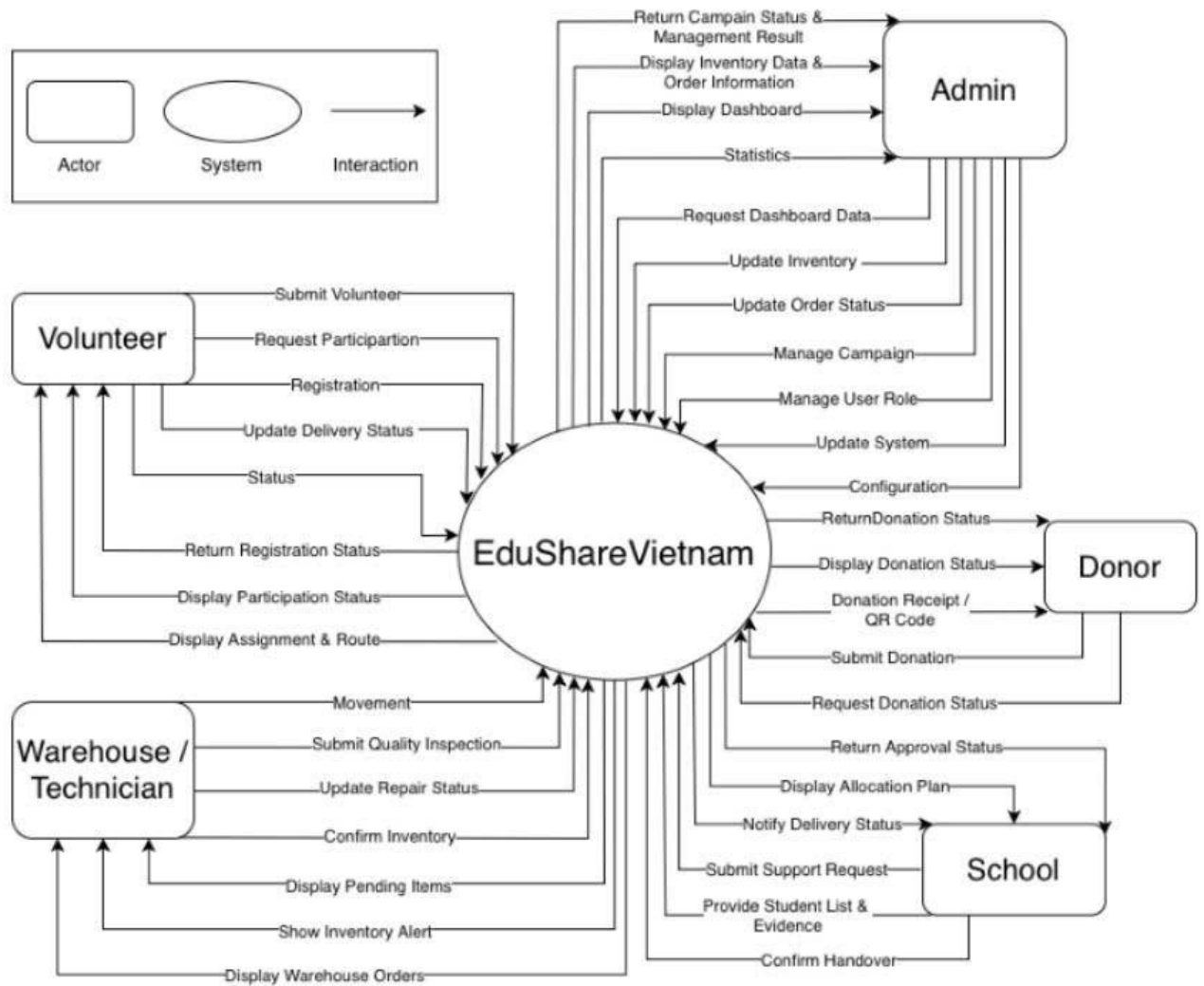


Figure 2. System Context Diagram

- Admin (Board of Administration / Coordinator):

- **Request Dashboard Data & Statistics:** Send request to extract data and statistics
Receive statistics, overview dashboard (Display Dashboard) of the entire system.
- **Update Inventory & Order Status:** Update the quantity of goods
inventory and change the status of export/import/dispatch orders. The following system
That will display the latest warehouse data and order information (Display
Inventory Data & Order Information).
- **Manage Campaign:** Manage and initiate donation and receipt campaigns
Campaign status and management results (Return Campaign Status &
Management Result).
- **Manage User Role:** Assign permissions, grant accounts and manage user roles
users in the system.
- **Update System Configuration:** Change and update configurations and parameters
system.

- Donor:

- **Submit Donation:** Submit registration information for donating educational resources
system.
- **Request Donation Status:** Query the progress of processing the donated item.
- **Receive Feedback & Status:** Receive donation status (Return Donation
Status / Display Donation Status) as well as electronic receipts and identifiers
QR of the item (Donation Receipt / QR Code).

- School (School / Beneficiary Representative):

- **Submit Support Request:** Submit a proposal request for resource support from the system
system.
- **Provide Student List & Evidence:** Provide a list of students who need support
and accompanying documents and photo evidence.
- **Receive Status & Allocation:** Receive notification about approval status (Return
Approval Status) and view detailed resource allocation plans from the coordination board
(Display Allocation Plan).

- **Delivery & Handover:** Receive shipping route notifications (Notify Delivery Status) and confirm delivery upon receipt of goods (Confirm Handover).
- **Warehouse / Technician (Storekeeper / Technician):**
 - **Movement & Confirm Inventory:** Update goods flows and confirm actual inventory quantity.
 - **Submit Quality Inspection:** Submit reports and quality inspection results for with the resources just received.
 - **Update Repair Status:** Update the refurbishment and repair status of devices technology (such as laptops, computers).
 - **Receive Alerts & Orders:** The system displays a list of available items pending items (Display Pending Items), warehouse import and export orders (Display Warehouse Orders) and issue warnings about warehouse status (Show Inventory Alert).
- **Volunteer (Volunteer):**
 - **Submit Volunteer Registration:** Submit application to register to participate in sex volunteer.
 - **Request & View Participation Status:** Request and view approval status hồ sơ tham gia (Return Registration Status / Display Participation Status).
 - **View Assignment & Route:** Get and display assigned tasks same travel route (Display Assignment & Route).
 - **Update Delivery Status:** Update delivery/receive status during the process transport.

13. Potential Risks and Mitigation Strategies

Table 9. *Risks and Mitigation*

Risk	Category	Impact	Mitigation Strategy
Donated items are of poor quality or cannot be used. Old damaged uniforms or outdated/severely damaged IT equipment that cannot be repaired, wasting transportation and warehousing costs.	Operations / Operations	Cao	•Require donors to upload actual photos and complete the self-declaration status list on the Donor Portal. •Implement the 3-level standard inspection process (Type A / Type B / Type C) right at the receiving stage. •Establish cooperation agreements with e-waste processors to recycle completely broken devices.
Data Disputes & Multi-Warehouse Inventory Conflicts (Race Conditions): Multiple coordinators simultaneously allocate the same shipment during a campaign peak ("Relay to School") leading to out of stock or deviation from actual quantity.	Technique	Cao	•Apply distributed locking mechanism (Distributed Locks) to hold inventory instantly. •Strictly enforce ACID integrity transactions with Optimistic Locking on PostgreSQL. •Record immutable audit logs for each step in changing the status of an artifact (Pending approval → Inspected → Ready to distribute → Released).
Leakage of privacy and Disadvantaged Student Data: Publicizing family information, images or identities of disadvantaged students violates regulations on personal data protection and children's rights.	Bảo mật / Pháp lý	Cao	•Perform Data Masking & Pseudonymization for all student data on public lookup portals. •Limit access to detailed records to authorized school representatives and administrators only. •Apply RBAC decentralization mechanism and encrypt all data transmitted via TLS 1.3 protocol.

Risk	Category	Impact	Mitigation Strategy
Fraud or exaggeration of support needs: Unverified organizations/individuals send fake lists of students to appropriate high-value devices (laptops, desktops, tablets).	Quản trị / Kiểm soát	Cao	•Required legal authentication of the school: required to upload a written request signed by the Board of Directors and stamped in red. •Implement the digital handover process (Proof of Delivery - PoD): take photos at the scene with GPS positioning, scan the QR code and sign electronically.
Volunteers absent or canceling shifts at the last minute: Volunteers registered but did not show up to do sorting duties at the warehouse or deliver goods during the peak period at the beginning of the school year.	Personnel operations	Average	•Build a list of backup volunteers (schedule 15-20% of the army). •Send automatic calendar reminders via Push Notification (Firebase FCM) and Email 24 hours before shift. •System integration Accumulate hours of dedication and issue an electronic Volunteer Certificate.
Supply and demand mismatch between geographical regions: There is too much donated goods in big cities (like Da Nang) while schools in remote areas (like the mountainous areas of Quang Nam) have a serious shortage.	Logistics / Coordination	Average	•Historically intelligent dispatch algorithm in Python to automatically propose internal transfer orders between nearby warehouses. •Update the shortage list in real time to the campaign homepage to direct the community to donate exactly what is needed.

Risk	Category	Impact	Mitigation Strategy
Interruption of network connection and Third Party Services: 3G/4G signal flickers in highland communes when making deliveries, or cloud storage service/notification of technical problems.	Technique	Average	•Design a React Native mobile application according to Offline-first architecture: store temporary data locally and automatically synchronize when there is a network connection. •Configure asynchronous processing queue (Task Queue) with automatic retry mechanism (Retry) when calling external API.
Project Completion progress multi-party business process (Receiving, Inspection, Repair, Warehouse Coordination, Volunteers, ESG Report) exceeding the 14–16 week time frame of the project.	Time	Average	•Strictly follow the Agile Scrum process with a 3-week/Sprint development cycle. •Prioritize completion of the core MVP modules (Donations, QR Code Assignment, Warehouse Management, Request Receiving, Coordination, PoD Handover, Dashboard).

14. Expected Outcomes / Deliverables

- **EduShare Vietnam Web/Application System:** Web and mobile application system Completely allows for comprehensive digitization of the process of receiving educational resources from home sponsorship, multi-point warehouse management with QR identification code, quality inspection, refurbishment computer repair, coordination of automatic allocation to disadvantaged schools and according to Transparently track the handover journey in real time.
- **Source Code:** Complete source code of the application (front-end and back-end) and system installation and configuration instructions.

- **Technical Documentation:**

- Proposal Document
- Project Plan Document
- Product Backlog Document
- User Stories Document
- Architecture Document
- Database Design
- UI Design
- Test Plan Document
- Test Case
- Sprint Backlog
- Code Standard
- Meeting Document
- Reflection Document

- **Final Report:**

- Comprehensive reports on the entire system research and development process.
- Analyze achieved results compared to initial goals.
- Evaluate limitations and suggest future directions for improvement, e.g
Such as AI applications that classify books and equipment through images, and AI that estimates usage value remaining, integrating Blockchain technology to make transaction logs transparent, and social impact analysis (ESG) dashboard for sponsoring businesses.

15. References

Table 10. References

No.	References	Document Information
1	Scrum Model	https://www.atlassian.com/agile/scrum
		https://www.digite.com/agile/scrum-methodology/
		https://docs.microsoft.com/en-us/azure/devops/boards/sprints/best-practices-scrum?view=azure-devops
		https://www.scrum.org/resources/scrum-guide
2	Technical	https://nodejs.org/en/learn/getting-started/introduction-to-nodejs
		https://aws.amazon.com/vi/what-is/api/
		https://microservices.io/
3.	Software Engineering Standards	https://www.nws.noaa.gov/oh/hrl/developers_docs/General_Software_Standards.pdf
		https://standards.ieee.org/standard/12208-2017.html
		https://sw-eng.larc.nasa.gov/

DESCRIPTION OF PRODUCT REQUIREMENTS"**16.1. Brief description of product ideas (less than 7 statements).****16.2. Requirements**

High-level functional requirements	1. Emergency Rescue Request Submission (SOS)
	2. GPS-based Location Detection and Map Display
	3. Rescue Provider / Volunteer Assignment
	4. Real-time Rescue Status Tracking
	5. User and Rescuer Account Management
	6. Notification and Alert System
	7. Rescue History and Reporting

Quality property requirements (e.g., related to issues: Easy to Use, Easy to Like, Easy to Learn, Easy to Understand, Easy to Buy / Yes,)	1. Easy to use
	2. Easy to learn
	3. Easy to interact
	4. Easy to understand
	5. Easy to monitor and analyze rescue statistics

Operational requirements (related to the following issues: Speed, Accuracy, Performance, Stability, Load Protection, Scalability, Safety, ...)	1. Ensure system stability and reliability
	2. Ensure accuracy of location and rescue information
	3. Ensure system security and data protection
	4. Support high concurrent access during emergency situations
	5. Ensure system scalability for future expansion

Environmental and operational requirements (related to the following issues: physical impact on the environment, interaction with relevant or existing systems, conditions of commercialization of products, ...)	1. Can be accessed on multiple web platforms (desktop, tablet, mobile browsers)
	2. Can integrate with external services such as map APIs and notification services
	3. Operates effectively under real-time conditions

Maintenance and support requirements	1. Easily maintain source code
	2. Easily expand source code
	3. Specific description of the document

Security/safety requirements (related to the following issues: conditions of use / access to products, personal freedom, inspection, ...)	1. User information and rescue data must be securely protected
	2. Secure authentication and authorization mechanisms
	3. Anti-intrusion and attack prevention mechanisms

Assess the complexity of technical problems		1. Regarding broad technical issues or contradictions
		2. No clear solution
		3. Resolve issues that are not covered by current standards and codes
	X	4. Relate to different stakeholder groups
	X	5. Include multiple component parts or sub-issues
	X	6. Related to many disciplines
	X	7. Have significant consequences in a wide range of context.

Standard requirements	X	1. Code standard.
	X	2. Design standards. (object-oriented design, analysis, and design templates...).
		3. IEEE (1058, 1540, 830, 1016, 829, 1012, 1008)
	X	4. ISO/IEC/IEEE 12207:2017 (TCVN 10539:2014); ISO/IEC 25051:2006 (TCVN 10540:2014);
		5. Other standards. (related to specific topics)